

DWT-DCT Frequency-Domain Forensic Watermarking

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Team: Team Laccha Paratha (ID: 138638) | Platform: TraceCrypt Offline Engine

ABSTRACT: Digital document distribution across untrusted channels demands forensic watermarking that survives lossy print-scan-photocopy pipelines while maintaining imperceptibility.

- 1. Methodology:** We evaluate a 2-level Discrete Wavelet Transform (DWT-2D) coupled with block Discrete Cosine Transform (DCT) mid-frequency spread-spectrum modulation. Reed-Solomon RS(32, 16) error correction coding provides robust forward recovery.
- 2. Fidelity Metrics:** Empirical testing confirms Peak Signal-to-Noise Ratio (PSNR) exceeding 44.2 dB and Structural Similarity Index (SSIM) above 0.98, rendering watermarked documents visually identical to original counterparts.
- 3. Experimental Results:** The extraction engine recovers complete 256-bit payloads under JPEG compression down to Quality 65 and Gaussian filtering without requiring access to original source documents.